docker_class_2(45)

- **Docker Service Initialization**
  Docker cannot be used unless the Docker service is running. Always start or restart the Docker service before executing Docker commands. It is also recommended to verify the Docker version before and after restarting the service.
- **Image Requirement for Container Creation**
  A Docker container cannot be created without a base image. The image acts as the template required to build and run a container.
- **Container Must Be Running for Access**
  Direct access to a container is not possible unless the container is in a running state. You must start the container first before executing commands or accessing it.
- **Running an Image Automatically Creates a Container**
  When a Docker image is executed using the `docker run` command, Docker automatically creates and starts a container from that image.

Component	Description
Docker Image	Template used to create containers
Docker Container	Running instance of a Docker image
docker run	Command that creates and starts a container

The **Docker Client** is the command-line interface used by users to interact with Docker.
The **Docker Host** is the system where Docker is installed and running.
The **Docker Daemon (`dockerd`)** runs on the Docker Host.
A **Docker Registry** is a storage location where Docker images are stored.

A **Docker Image** is a template, and a **Docker Container** is the running instance created from that template.

launch ec2 and install docker there
yum install docker -y
check docker status
docker --version
start docekr
docker status
docker info: to get full information
Docker Client sends commands, and Docker Host executes them by creating and managing containers.

docker images

to get images where you can get images ?? docker hub

ex: docker pull ubuntu

it is download you can check

docker images

every image have the id it is created a 4 weeks ago

Now we will create containers

to check container is it there or not i.e

docker ps

to create a container

docker run -it --name flm-container ubuntu

i=interactive terminal

t=terminal

it= it will create shell inside the container which is used to perform commands or excute programms

--name we have to give container name so it is

```
[root@ip-172-31-22-105 ~]# docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[root@ip-172-31-22-105 ~]# docker run -it --name flm-container  ubuntu
root@4641be85ef66:/#
```

so we enter inside container if you give ls you can see files and folder

if you want to come out side the container just exit

now you will be become

if you do docker ps -a

```
exit
[root@ip-172-31-22-105 ~]# docker ps -a
CONTAINER ID   IMAGE     COMMAND       CREATED          STATUS                     PORTS     NAMES
4641be85ef66   ubuntu    "/bin/bash"   56 seconds ago   Exited (0) 8 seconds ago             flm-container
[root@ip-172-31-22-105 ~]#
```

it will show status and command mode and we are seeing exited status we are seeing 8 sec ago

if we want to go inside container give docker attach flm-container

you can not enter the container it should start the container to start the container

docker start flm-container

docker ps -a

now check docker status you will see up status

```
[root@ip-172-31-22-105 ~]# docker ps -a
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS         PORTS     NAMES
4641be85ef66   ubuntu    "/bin/bash"   2 minutes ago   Up 4 seconds             flm-container
[root@ip-172-31-22-105 ~]# docker attach  flm-container
root@4641be85ef66:/#
```

now docker attach flm-container you can

now we are in container

```
[root@ip-172-31-22-105 ~]# docker ps -a
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS                     PORTS     NAMES
4641be85ef66   ubuntu    "/bin/bash"   3 minutes ago   Exited (0) 3 seconds ago             flm-container
[root@ip-172-31-22-105 ~]#
```

if we do exit then again we can see exit mode

if we see like that the application will run no it will not run

again if you start container and enter the container you are inside the container then this time iam doing control + pq
you will get outside but container will run
you donot see this time exit mode

Running containers docker ps
All containers (running + stopped) docker ps -a
Only exited containers to see only stopping container
Only exited containers
docker ps -f "status=exited"
how to delete container docker rm containerid/containername
Note: running container we can not remove before remove we have to stop the container
to stop all the container at a time: docker stop $(docker ps)
to start all the container at a time: docker start $(docker ps -a)
to rename the container: docker rename cont-1 cont-1111
it will kill/exit with early as compaired to stop docker kill containername
stop will take some time to exit

to see list containers all: docker ps or docekr container ls
to see latest 2 containers: docker ps -n 2 or docker container ls -a -n 2
to see latest docker container ls --latest
it will delete exited container all at a time: docker container rm $(docker container ls -a)
to see the size of the container: docker container ls -s
Check docker memory and cpu usage: docker stats

Fix

- restart container
- optimize app
- set limits:

docker run -m 512m ...